Two spins coupled to an XXZ bulk

Reduced dynamics and quantum-state texture

Spin transport project

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Abstract

We study the joint reduced dynamics of two spin-1/2 probes, each attached to one end of a finite XXZ bulk. The total Hamiltonian generates unitary evolution, and the initial probe–bulk state is a product. For Ising boundary coupling, four conditional bulk Hamiltonians give an exact time-dependent Schur, or generalized dephasing, channel. Its complete action is encoded in the output from the single probe preparation $|{+}{+}\rangle$. For exchange coupling, the available description gives the symmetry blocks rather than closed expressions for their coefficients. When the Hamiltonian conserves total magnetization and the bulk preparation commutes with bulk magnetization, the reduced superoperator and Choi matrix split into blocks of sizes $1, 4, 6, 4, 1$. Quantum-state texture is evaluated only on the output from $|{+}{+}\rangle$; exact relations to the Ising channel, its fidelities and its entanglement-generating ability follow from the Schur structure.

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1 Introduction

Two probes coupled to a common spin chain form an open quantum system whose reduced dynamics depends on the interaction at each contact. We consider Ising and exchange contacts separately, with no direct probe–probe interaction. The bulk starts in a fixed state, uncorrelated with the probe pair. Tracing out the bulk after unitary evolution defines a quantum channel on the pair.

Ising contacts conserve both probe z projections and produce generalized dephasing in the joint computational basis. Exchange contacts allow spin transfer between the probes and the bulk. Conservation of total magnetization then constrains the reduced channel when the bulk preparation respects the same rotation symmetry.

These two interactions lead to the following channel descriptions:

1. Ising coupling: an exact conditional-Hamiltonian reduction, an exact Schur channel, and reconstruction of that channel from the full output state for $|{+}{+}\rangle$.
2. Exchange coupling: exact total-magnetization sectors and, for a magnetization-invariant bulk preparation, a phase-covariant channel with five coherence-order blocks. These constraints leave the block entries to be computed from the many-body evolution.
3. Texture: exact grand-sum, return-probability and response-operator identities. For Ising coupling these yield channel fidelities and other properties without additional input preparations.

The derivations use exact finite-system dynamics, without a weak-coupling or Markov approximation. The channel structure fixes which probe operators can mix; their time dependence depends on the bulk Hamiltonian and its initial state.

2 Model, conventions and the reduced channel

2.1 Geometry and Hamiltonians

Let $N \geq 2$ be the number of *bulk* spins. There are $N + 2$ spins in total, with physical order $A, 1, \ldots, N, B$. We use

$$S^\alpha = \sigma^\alpha/2, \qquad S^\pm = S^x \pm iS^y = (\sigma^x \pm i\sigma^y)/2, \qquad \hbar = 1. \tag{1}$$

The basis convention is $|0\rangle = |\uparrow\rangle$, $|1\rangle = |\downarrow\rangle$, and the probe basis is $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. For formulas involving a product of probe and bulk operators, we group the Hilbert space as $\mathcal{H}_{AB} \otimes \mathcal{H}_E$ by a fixed permutation of the physical site ordering.

The reference bulk and the two alternative interactions are

$$H_E = J \sum_{j=1}^{N-1} (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z), \tag{2}$$

$$V_{\mathrm{ZZ}} = g_A S_A^z S_1^z + g_B S_N^z S_B^z, \tag{3}$$

$$\begin{aligned} V_{\mathrm{ex}} &= g_A (S_A^x S_1^x + S_A^y S_1^y) + g_B (S_N^x S_B^x + S_N^y S_B^y) \\ &= \frac{g_A}{2} (S_A^+ S_1^- + S_A^- S_1^+) + \frac{g_B}{2} (S_N^+ S_B^- + S_N^- S_B^+). \end{aligned} \tag{4}$$

Thus $H^{(r)} = H_E + V_r$, with $r = \mathrm{ZZ}$ or ex. Identity operators on the remaining spins are implicit. The parameters J, g_A, g_B have energy units and Δ is dimensionless. Equal couplings mean $g_A = g_B = g$. The exchange flip-flop matrix element at the left contact is $g_A/2$ in this convention.

Equation (2) has no applied fields. Longitudinal bulk or probe fields preserve the magnetization arguments below, with their phases included in the conditional dynamics. Transverse *bulk* fields also preserve the Ising Schur reduction, but generally invalidate the exchange magnetization reduction because they break conservation of total z magnetization.

2.2 Product preparation and exact channel

Fix any bulk density matrix ω_E, independent of the probe input. The initial state and reduced evolution are

$$\rho(0) = \rho_{AB}(0) \otimes \omega_E, \tag{5}$$

$$U_t = e^{-iH^{(r)}t}, \qquad \Lambda_t(X) = \mathrm{Tr}_E[U_t(X \otimes \omega_E)U_t^\dagger], \tag{6}$$

$$\rho_{AB}(t) = \Lambda_t(\rho_{AB}(0)). \tag{7}$$

The product assumption separates the pair from the bulk; it imposes no factorization between A and B for the definition of the channel. Only the texture protocol fixes their input to $|{+}{+}\rangle$.

To see complete positivity and trace preservation directly, write $\omega_E = \sum_\nu p_\nu |\phi_\nu\rangle\langle\phi_\nu|$ and choose any orthonormal bulk output basis $\{|r\rangle\}$. Then

$$K_{r\nu}(t) = \sqrt{p_\nu}\, {}_E\langle r|U_t|\phi_\nu\rangle_E, \quad \Lambda_t(X) = \sum_{r,\nu} K_{r\nu} X K_{r\nu}^\dagger, \quad \sum_{r,\nu} K_{r\nu}^\dagger K_{r\nu} = I_4. \tag{8}$$

These are formal exact expressions in both regimes; evaluating them for an interacting bulk remains a many-body problem. Correlated probe–bulk preparations have reduced trajectories, but do not in general define the same CPTP map on arbitrary probe inputs.

2.3 Choi normalization

The *normalized* Choi state, with the output factor first, is

$$|\Phi_4\rangle = \frac{1}{2} \sum_{a=1}^{4} |a\rangle_{AB} |a\rangle_R, \qquad J_t = (\Lambda_t \otimes \mathrm{id}_R)(|\Phi_4\rangle\langle\Phi_4|). \tag{9}$$

It obeys $J_t \succeq 0$, $\mathrm{Tr}\, J_t = 1$ and $\mathrm{Tr}_{AB}\, J_t = I_R/4$. The corresponding unnormalized Choi matrix is $C_t = 4J_t$. The auxiliary system R represents a reference for the channel and is independent of the physical bulk.

3 Ising coupling: exact generalized dephasing

3.1 Four conditional Hamiltonians

Since $[H^{(\mathrm{ZZ})}, S_A^z] = [H^{(\mathrm{ZZ})}, S_B^z] = 0$, label each probe basis state by $a = (s_A, s_B)$ with $s_A, s_B = \pm 1/2$, and set $P_a = |a\rangle\langle a|$. The Hamiltonian and propagator are

$$H^{(\mathrm{ZZ})} = \sum_a P_a \otimes H_a, \qquad H_a = H_E + g_A s_A S_1^z + g_B s_B S_N^z, \tag{10}$$

$$U_t = \sum_a P_a \otimes U_a(t), \qquad U_a(t) = e^{-iH_a t}. \tag{11}$$

This follows by inserting the spectral decompositions of S_A^z, S_B^z; orthogonal projectors exponentiate independently. It is exact for any bulk Hamiltonian acting only on E, including one with transverse fields.

For equal couplings, the four perturbations are

a	$H_a - H_E$
00	$\frac{g}{2}(S_1^z + S_N^z)$
01	$\frac{g}{2}(S_1^z - S_N^z)$
10	$-\frac{g}{2}(S_1^z - S_N^z)$
11	$-\frac{g}{2}(S_1^z + S_N^z)$

. (12)

These configurations label invariant branches. The input may still coherently occupy all four branches.

3.2 Schur form and its physical constraints

Let $E_{ab} = |a\rangle\langle b|$. Since $P_c E_{ab} P_d = \delta_{ca}\delta_{db} E_{ab}$, tracing the controlled propagator gives

$$\boxed{\Lambda_t(E_{ab}) = \Gamma_{ab}(t) E_{ab}, \qquad \Lambda_t(X) = \Gamma(t) \circ X,} \tag{13}$$

where $\circ$ is the entrywise Schur product and

$$\Gamma_{ab}(t) = \mathrm{Tr}_E[U_a(t)\omega_E U_b(t)^\dagger]. \tag{14}$$

For a pure bulk, this is the complex conditional overlap $\langle\phi_0|U_b^\dagger U_a|\phi_0\rangle$. Its squared modulus is a conditional Loschmidt echo; the complex amplitude, including its phase, is needed for the channel.

The matrix Γ is a correlation matrix:

$$\Gamma = \Gamma^\dagger \succeq 0, \qquad \Gamma_{aa} = 1, \qquad |\Gamma_{ab}| \le 1, \qquad \Gamma_{ab}(0) = 1. \tag{15}$$

Indeed, setting $k_{r\nu,a} = \sqrt{p_\nu}\langle r|U_a|\phi_\nu\rangle$ gives $\Gamma_{ab} = \sum_{r,\nu} k_{r\nu,a} k^*_{r\nu,b}$, a positive Gram factorization. The unit diagonal follows from unitarity and $\mathrm{Tr}\,\omega_E = 1$; 2×2 principal minors give the modulus bound. Conversely, every positive matrix with unit diagonal defines a CPTP Schur map. This converse characterizes the map class, not which members a specified XXZ Hamiltonian realizes at a given time.

All four populations are fixed and $\Lambda_t(I_4) = I_4$. Six complex off-diagonal factors specify the nontrivial action, subject to positivity. The common interacting bulk can correlate the two probes, so a product of independent single-probe dephasing channels cannot be assumed. There is no general monotonicity of $|\Gamma_{ab}(t)|$.

3.3 Dephasing and depolarization

A depolarizing map has the form $\mathcal{D}_\lambda(X) = \lambda X + (1-\lambda)\,\mathrm{Tr}(X) I_4/4$. The Ising map fixes every P_a. Requiring $\mathcal{D}_\lambda(P_a) = P_a$ forces $\lambda = 1$. Thus a nonidentity Ising Schur map cannot be a depolarizing map. Dephasing includes coherent diagonal rotations and revivals; it does not imply irreversible or strictly dissipative evolution.

3.4 One output determines the complete channel

The sole probe preparation used for texture is

$$|f_1\rangle = |+\rangle_A \otimes |+\rangle_B = \frac{1}{2}\sum_{a=1}^{4}|a\rangle, \qquad f_1 = |f_1\rangle\langle f_1|, \qquad (f_1)_{ab} = \frac{1}{4}. \tag{16}$$

Denote its full output state by $\rho_f(t) = \Lambda_t(f_1)$. Then

$$\boxed{\rho_f(t) = \Gamma(t)/4, \qquad \Gamma(t) = 4\rho_f(t), \qquad \Lambda_t(X) = [4\rho_f(t)] \circ X.} \tag{17}$$

Consequently the *entire density matrix* of one known output determines every property of the instantaneous Ising channel. Temporal properties require these outputs at the relevant times. A single scalar texture value cannot replace this matrix.

The populations of ρ_f are $1/4$, so its 12 remaining real parameters are its six complex coherences. In a Pauli expansion, $\langle ZI\rangle = \langle IZ\rangle = \langle ZZ\rangle = 0$. The eight local measurement settings $(\alpha,\beta) \neq (z,z)$, $\alpha,\beta \in \{x,y,z\}$, recover the other Pauli coefficients, including single-spin expectations from their marginals. Thus one input preparation together with output tomography suffices.

3.5 Choi, Kraus and spectral consequences

Let $W : \mathcal{H}_{AB} \to \mathcal{H}_{AB} \otimes \mathcal{H}_R$ be the isometry $W|a\rangle = |aa\rangle$. Equations (9) and (13) give

$$J_t = \frac{1}{4}\sum_{a,b}\Gamma_{ab}|aa\rangle\langle bb| = W\rho_f(t)W^\dagger. \tag{18}$$

The Choi support has dimension at most four. In particular,

$$\mathrm{rank}\, J_t = \mathrm{rank}\,\Gamma \leq 4, \tag{19}$$

$$\mathrm{spec}(J_t) = \mathrm{spec}(\Gamma/4) \text{ with twelve additional zeros}, \tag{20}$$

$$S(J_t) = S(\rho_f), \qquad \mathrm{Tr}\, J_t^2 = \mathrm{Tr}\,\rho_f^2 = \frac{1}{16}\sum_{a,b}|\Gamma_{ab}|^2. \tag{21}$$

If $\Gamma = \sum_k \lambda_k v_k v_k^\dagger$ with normalized eigenvectors, one minimal Kraus family is

$$D_k = \sqrt{\lambda_k}\ \mathrm{diag}(v_{k,1},\ldots,v_{k,4}), \qquad \Lambda_t(X) = \sum_k D_k X D_k^\dagger. \tag{22}$$

The superoperator is diagonal in the E_{ab} basis, with eigenvalues Γ_{ab} and singular values $|\Gamma_{ab}|$. Choi rank and superoperator rank are distinct: the identity has Choi rank one and superoperator rank sixteen.

The map is unitary iff $\mathrm{rank}\,\Gamma = 1$, in which case $\Gamma_{ab} = u_a u_b^*$ with $|u_a| = 1$, giving the diagonal unitary $\mathrm{diag}(u_1,\ldots,u_4)$. This does not require that the unitary factor into two local rotations.

The partial transpose of J_t has eigenvalues $1/4$ (four times) and $\pm|\Gamma_{ab}|/4$ for each $a < b$. Hence its reference–output negativity is

$$\mathcal{N}_{AB|R}(J_t) = \frac{1}{4}\sum_{a<b}|\Gamma_{ab}|. \tag{23}$$

This is distinct from physical $A|B$ entanglement. The Ising map is entanglement breaking iff all off-diagonal Γ_{ab} vanish: otherwise its Choi state has a negative partial transpose, while the diagonal case has the explicitly separable Choi state $\frac{1}{4}\sum_a |aa\rangle\langle aa|$.

3.6 Time-local dynamics and intermediate maps

The map is invertible iff every $\Gamma_{ab}(t)$ is nonzero. At regular times its exact time-local generator obeys

$$\mathcal{L}_t(E_{ab}) = \frac{\dot{\Gamma}_{ab}(t)}{\Gamma_{ab}(t)}E_{ab}. \tag{24}$$

If Λ_s is invertible, the unique intermediate linear map is Schur multiplication by $Q_{ab}(t,s) = \Gamma_{ab}(t)/\Gamma_{ab}(s)$. Its diagonal is one, so

$$\Lambda_t\Lambda_s^{-1} \text{ is CPTP} \quad\Longleftrightarrow\quad Q(t,s) \succeq 0. \tag{25}$$

At singular times the division is undefined and this test cannot be applied. A vanished coherence factor that later becomes nonzero already precludes divisibility through that time, since a linear intermediate map cannot recover the annihilated matrix unit.

3.7 Probe entanglement from the same input

For two-qubit states, concurrence C and negativity $\mathcal{N}$ test entanglement of A with B; their definitions are in Appendix A. For a Schur channel, the output from $|{+}{+}\rangle$ also determines the maximum concurrence generated from separable probe states.

Proposition 3.1. *For a fixed Ising Schur channel, its maximum output concurrence over separable probe inputs is*

$$\max_{\rho\in\mathrm{Sep}(A:B)} C[\Lambda_t(\rho)] = C[\rho_f(t)] = C[\Gamma(t)/4]. \tag{26}$$

Proof. For a normalized pure product input $(a_0|0\rangle{+}a_1|1\rangle)\otimes(b_0|0\rangle{+}b_1|1\rangle)$, set $A = \sqrt{2}\,\mathrm{diag}(a_0,a_1)$ and $B = \sqrt{2}\,\mathrm{diag}(b_0,b_1)$. The output is $(A\otimes B)\rho_f(A\otimes B)^\dagger$ and has unit trace because ρ_f has uniform diagonal. The local-filter identity for two-qubit concurrence gives $C_{\mathrm{out}} = |\det A\det B|C(\rho_f) = 4|a_0a_1b_0b_1|C(\rho_f) \le C(\rho_f)$. The identity follows from the spin-flip formula; singular filters follow by continuity. Convexity extends the bound to mixtures of product inputs. The input $|{+}{+}\rangle$ attains equality. □

In particular, the Ising map can create probe entanglement from some separable input iff $\rho_f(t)$ is entangled. Texture and Choi reference–output negativity are different quantities and do not replace this criterion.

4 Exchange coupling: exact symmetry reduction

4.1 Total charge and Hamiltonian sectors

For the field-free model define

$$M_S = S_A^z + S_B^z, \qquad M_E = \sum_{j=1}^{N} S_j^z, \qquad M = M_S + M_E. \tag{27}$$

Each exchange term transfers one unit between a probe and the bulk, so $[H^{(\mathrm{ex})}, M] = 0$, while individual probe S^z values generally are not conserved. The total-charge sector m has dimension

$$\dim \mathcal{H}_m = \binom{N+2}{(N+2)/2 - m}. \tag{28}$$

This Hamiltonian decomposition requires no assumption on the input; coherences between occupied sectors must still be retained. There is no four-branch controlled propagator of the form (10) for exchange coupling.

4.2 Bulk assumption and proof of phase covariance

For the channel reduction assume

$$[\omega_E, M_E] = 0. \tag{29}$$

Support in a single well-defined bulk magnetization sector is sufficient. An incoherent mixture of sectors also suffices. A coherent superposition of different bulk magnetizations generally does not.

Let $V_\theta = e^{-i\theta M_S}$ and $W_\theta = e^{-i\theta M_E}$. Conservation gives $[U_t, V_\theta \otimes W_\theta] = 0$, while (29) gives $W_\theta \omega_E W_\theta^\dagger = \omega_E$. Inserting these into the partial trace proves

$$\boxed{\Lambda_t(V_\theta X V_\theta^\dagger) = V_\theta \Lambda_t(X) V_\theta^\dagger.} \tag{30}$$

This is $U(1)$ phase covariance. It permits exchange of magnetization with the bulk; it does not imply $\Lambda_t^*(M_S) = M_S$ or unitality.

4.3 Five superoperator blocks

The probe charges are $(m_{00}, m_{01}, m_{10}, m_{11}) = (1, 0, 0, -1)$. The operator E_{ab} has coherence order $q = m_a - m_b$, meaning $[M_S, E_{ab}] = qE_{ab}$. Under a rotation it acquires the phase $e^{-iq\theta}$, so covariance implies

$$\Lambda_t(\mathcal{V}_q) \subseteq \mathcal{V}_q, \qquad \mathcal{V}_q = \mathrm{span}\{E_{ab} : m_a - m_b = q\}. \tag{31}$$

The ordered bases $\mathcal{B}_q$ of these spaces are

q	Dimension	Basis of $\mathcal{V}_q$
0	6	$E_{00,00}, E_{01,01}, E_{10,10}, E_{11,11}, E_{01,10}, E_{10,01}$
+1	4	$E_{00,01}, E_{00,10}, E_{01,11}, E_{10,11}$
−1	4	Adjoints of the $q = +1$ basis
+2	1	$E_{00,11}$
−2	1	$E_{11,00}$

Thus the 16×16 superoperator has the exact reduction

$$\boxed{\Lambda_t \simeq T^{(-2)} \oplus T^{(-1)} \oplus T^{(0)} \oplus T^{(+1)} \oplus T^{(+2)}, \quad (d_q) = (1, 4, 6, 4, 1).} \tag{32}$$

Operators of the same order may mix. Hermiticity relates the opposite orders. In the ordered basis $\mathcal{B}_q$, the exact entries are

$$T^{(q)}_{cd,ab}(t) = \sum_{r,\nu} (K_{r\nu})_{ca} (K_{r\nu})^*_{db}, \qquad E_{ab}, E_{cd} \in \mathcal{B}_q. \tag{33}$$

Equation (33) is a formal construction, not a closed evaluation of these functions for general interacting XXZ dynamics.

The normalized Choi state satisfies the equivalent symmetry condition

$$[J_t, M_S \otimes I_R - I_{AB} \otimes M_S^{\mathsf{T}}] = 0. \tag{34}$$

Grouping output–reference kets $|c\rangle|a\rangle$ by $m_c - m_a$ gives Choi blocks of the same sizes $1, 4, 6, 4, 1$. Choi blocks and superoperator blocks are different rearrangements of channel coefficients and should not be identified as matrices or assigned the same spectra.

4.4 Kraus operators with definite charge transfer

Under (29), diagonalize ω_E within its charge sectors: $M_E|\phi_\nu\rangle = \mu_\nu|\phi_\nu\rangle$. Choose $M_E|r\rangle = \mu_r|r\rangle$. Conservation of the total input and output charge in Eq. (8) yields

$$(K_{r\nu})_{ca} \neq 0 \implies m_c - m_a = \mu_\nu - \mu_r, \qquad [M_S, K_{r\nu}] = (\mu_\nu - \mu_r) K_{r\nu}. \tag{35}$$

Only transfers between -2 and $+2$ can act on the two probes. The transfer label belongs to a Kraus operator; coherence order labels an input or output matrix unit. Their roles in the Choi and superoperator reductions are related but distinct.

4.5 Reduced states at fixed total magnetization

If the *global* initial state is supported in one total-M sector, the reduced probe state has the exact form

$$\rho_{AB}(t) = \begin{pmatrix} p_{00} & 0 & 0 & 0 \\ 0 & p_{01} & c & 0 \\ 0 & c^* & p_{10} & 0 \\ 0 & 0 & 0 & p_{11} \end{pmatrix}. \tag{36}$$

For a pure global ket, resolve its four probe components. Their bulk kets have charges $m - 1, m, m, m + 1$. Orthogonality of different bulk charges eliminates every coherence except $01 \leftrightarrow 10$. Linearity extends the conclusion to mixtures within the sector. The populations and c remain time dependent, and the concurrence is

$$C = 2 \max\{0, |c| - \sqrt{p_{00} p_{11}}\}. \tag{37}$$

This restriction does *not* apply to the texture protocol merely because the bulk has fixed magnetization. The input $|{+}{+}\rangle$ contains probe charges $+1, 0, -1$ with probabilities $1/4, 1/2, 1/4$, and hence occupies three total-charge sectors for a fixed-charge bulk. Their coherences cannot be discarded. In particular, f_1 itself is not of the form (36).

Reflection and global spin flip give additional covariance only when both the Hamiltonian and bulk preparation have the required symmetry. Equal contacts or a symmetric Hamiltonian alone do not impose that symmetry on an arbitrary reduced trajectory.

5 Quantum-state texture of the $|{+}{+}\rangle$ preparation

5.1 Grand sum and directly measurable return probability

Texture is defined relative to the probe computational basis, whose textureless state is f_1 in Eq. (16). For any probe state, the grand sum is real and satisfies

$$\Sigma(\rho) = \sum_{a,b} \rho_{ab} = 4\langle f_1|\rho|f_1\rangle, \qquad 0 \le \Sigma(\rho) \le 4. \tag{38}$$

For the single physical input $f_1 \otimes \omega_E$, set

$$p_f(t) = \langle f_1|\rho_f(t)|f_1\rangle = \Sigma(\rho_f(t))/4, \quad T_F(\rho_f) = 1 - p_f, \quad R(\rho_f) = -\ln p_f. \tag{39}$$

These are the fidelity texture and rugosity conventions used here [1, 2]. At $p_f = 0$, $R = +\infty$; initially $p_f = 1$ and both textures vanish. The texturizing powers evaluated on the unique textureless input are $\Omega_{T_F}(t) = T_F[\Lambda_t(f_1)]$ and $\Omega_R(t) = R[\Lambda_t(f_1)]$.

Since $4f_1 = (I + \sigma_A^x)(I + \sigma_B^x)$,

$$\Sigma(\rho_f) = 1 + \langle\sigma_A^x\rangle + \langle\sigma_B^x\rangle + \langle\sigma_A^x\sigma_B^x\rangle. \tag{40}$$

A joint x-basis measurement gives $p_f = P_{++}^{(x)}$ directly. Equivalently, apply a Hadamard to each probe and record outcome 00. This uses one measurement setting, whereas full channel reconstruction in the Ising case uses the additional coherence measurements described in Section 3.4.

5.2 Ising texture and channel fidelities

Using $\rho_f = \Gamma/4$ gives

$$p_f = \frac{1}{16}\sum_{a,b}\Gamma_{ab} = \frac{1}{4} + \frac{1}{8}\sum_{a<b}\Re\Gamma_{ab}, \tag{41}$$

$$\Omega_{T_F} = \frac{3}{4} - \frac{1}{8}\sum_{a<b}\Re\Gamma_{ab}, \qquad \Omega_R = -\ln\left(\frac{1}{16}\sum_{a,b}\Gamma_{ab}\right). \tag{42}$$

For the same Schur channel, the entanglement fidelity relative to the identity is $F_e = \langle\Phi_4|J_t|\Phi_4\rangle = p_f$. Therefore the average pure-input fidelity relative to the identity is

$$\boxed{F_e = p_f = \Sigma(\rho_f)/4, \qquad F_{\text{avg}} = \frac{4F_e + 1}{5} = \frac{1 + \Sigma(\rho_f)}{5}.} \tag{43}$$

The general relation between average and entanglement fidelities is standard [3]; $F_e = p_f$ uses the Schur structure. These are fidelities of the *two-probe map relative to the identity*, not fidelities of sending an unknown state from A to B.

Only the real sum of the six coherences enters p_f. It does not fix their individual magnitudes or phases and does not determine purity, probe entanglement, or CP divisibility. Coherent phase evolution can change texture even when the map is unitary.

5.3 Response operator and texture preservation

The dual channel gives

$$\mathcal{R}_t = 4\Lambda_t^*(f_1), \qquad \Sigma[\Lambda_t(X)] = \mathrm{Tr}(\mathcal{R}_t X). \tag{44}$$

In the Ising case, comparing coefficients of X_{ab} yields

$$\mathcal{R}_t = \Gamma(t)^{\mathsf{T}}. \tag{45}$$

The transpose is ordinary, not a Hermitian conjugate. In general, fixing the free state, $\Lambda_t(f_1) = f_1$, and preserving the grand sum for every input, $\Lambda_t^*(f_1) = f_1$, are different conditions [2]. For this Schur family each condition forces $\Gamma_{ab} = 1$ for all a, b, hence the identity channel. Consequently $\Omega_{T_F} = 0$ also certifies the identity within the known Ising family.

5.4 Exchange texture and the limits of one input

Equations (39), (40) and (44) hold for exchange coupling as well. However, ρ_f is generally not a Schur multiplier matrix and does not determine the exchange channel. Under Eq. (29), f_1 has a nonzero component in every $\mathcal{V}_q$, but one output determines only each block's action on that particular component. It cannot determine a general 6×6 or 4×4 block.

There is one exact scalar exception. Since $\mathcal{V}_{+2}$ is one-dimensional, write $\Lambda_t(E_{00,11}) = \eta_{+2}(t) E_{00,11}$. Then

$$\eta_{+2}(t) = 4[\rho_f(t)]_{00,11}, \qquad \eta_{-2}(t) = \eta_{+2}(t)^*. \tag{46}$$

This inference requires covariance; it is not guaranteed for a bulk coherent across magnetization sectors or for transverse-field dynamics. The equality $F_e = p_f$ is likewise not a general exchange-channel identity, although $F_{\text{avg}} = (4F_e + 1)/5$ still holds.

6 Channel structure and dynamical coefficients

Six complex overlaps between conditional bulk evolutions determine the Ising channel. The output $\rho_f = \Gamma/4$ therefore specifies its full action, including its Choi state, fidelities and ability to generate probe entanglement. Evaluating the overlaps requires the conditional dynamics for the chosen J, Δ, g_A, g_B, N and ω_E.

For exchange coupling, conservation of total magnetization and a magnetization-invariant bulk preparation give five coherence-order blocks. Symmetry fixes their dimensions and forbids mixing between different orders, but leaves the entries within each block to be calculated. The output from $|{+}{+}\rangle$ determines the texture and the one-dimensional highest-order response; it does not specify the full exchange channel. A general closed expression for the interacting exchange coefficients remains unavailable.

A State diagnostics and their interpretation

For $\rho = \rho_{AB}$, the state purity and entropy are

$$P = \operatorname{Tr} \rho^2, \qquad S(\rho) = -\sum_k \lambda_k \ln \lambda_k, \qquad 0 \ln 0 = 0. \tag{47}$$

Logarithms are natural. The probe mutual information is $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho)$, with $\rho_A = \operatorname{Tr}_B \rho$ and $\rho_B = \operatorname{Tr}_A \rho$. It includes classical and quantum correlations.

Probe negativity and concurrence are

$$\mathcal{N}_{A|B}(\rho) = \frac{\|\rho^{T_B}\|_1 - 1}{2}, \tag{48}$$

$$\widetilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \qquad C(\rho) = \max(0, r_1 - r_2 - r_3 - r_4), \tag{49}$$

where the r_i are the decreasing nonnegative square roots of the eigenvalues of $\rho\widetilde{\rho}$ [4]. For two qubits, $C > 0$ and $\mathcal{N} > 0$ are equivalent tests of entanglement. They concern $A|B$, while $S(\rho_{AB})$ concerns the pair's marginal mixedness. Only for a globally pure state does $S(\rho_{AB})$ equal entanglement entropy across $AB|E$.

For the Ising texture input,

$$P_f(t) = \frac{1}{4} + \frac{1}{8}\sum_{a<b} |\Gamma_{ab}(t)|^2, \qquad \frac{1}{4} \le P_f \le 1. \tag{50}$$

Its local probe coherences are

$$(\rho_A)_{01} = \frac{\Gamma_{00,10} + \Gamma_{01,11}}{4}, \qquad (\rho_B)_{01} = \frac{\Gamma_{00,01} + \Gamma_{10,11}}{4}, \tag{51}$$

and each local diagonal is $(1/2, 1/2)$. For a pure bulk, the global state is $\frac{1}{2}\sum_a |a\rangle \otimes U_a|\phi_0\rangle$; it is a product across $AB|E$ iff its four conditional bulk kets are proportional. For a mixed bulk, reduced entropy or purity alone does not quantify entanglement across that cut.

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